

Monday Warm-Up: Unit 3, Magnetic Forces and Fields

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1 Memory Bank

- $\vec{F} = I\vec{L} \times \vec{B}$... The Lorentz force on a current.
- $\vec{F} = I\vec{L} \times \vec{B}$... Lorentz Force on a Current
- $\vec{B} = (\mu_0 I)/(2\pi r)\hat{\phi}$... The magnetic field B caused by the current I a distance r away. The field *circles* the wire, so the direction of $\vec{B}$ is $\hat{\phi}$.
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$.
- $\vec{B} = \mu_0 n I \hat{z}$... The B-field inside a solenoid. $n = N/L$, the turns per unit length.
- $\vec{B} = \mu_0 I/(2R)\hat{z}$... The B-field inside a circular loop with radius R .

the resistance of the solenoid is 5Ω , what is the power needed to create this B-field?

2 Force on an Element of Current

1. A wire on a circuit board has a length $L=4.7\text{mm}$ and carries a current $I=2.23 \mu\text{A}$ in the positive y direction. A nearby circuit element generates a magnetic field in the vicinity of the wire given by

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k} \quad (1)$$

with $B_x = 5 \text{ G}$, $B_y = 4 \text{ G}$, and $B_z = 3 \text{ G}$. (a) Calculate the force on the current element due to the field $\vec{B}$. (b) Compare the magnitude of this result to the maximum possible force on the current, ILB .

2. To see why an MRI utilizes iron to increase the magnetic field created by a coil, calculate the current needed in a 400-loop-per-meter circular coil 0.660 m in radius to create a 1.20-T field (typical of an MRI instrument) at its center with no iron present. The magnetic field of a proton is approximately like that of a circular current loop $0.650 \times 10^{-15} \text{ m}$ in radius carrying $1.05 \times 10^4 \text{ A}$. What is the field at the center of such a loop?

3 Ampère's Law

1. Consider a long, closely wound solenoid with $5,000$ turns per meter. (a) What current is needed to produce a magnetic field inside the solenoid of 10^{-4} T ? (b) If